Research Article

Induced Paths, Radius, and Extremal Neighbourhood Independence in Graphs without 4-Cycles

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Abstract

Every connected graph with no 4-cycle subgraph and no induced path on seven vertices satisfies $S \le 3n$, where S is the sum of the independence numbers of the open neighbourhoods, with equality only for the Petersen graph. Every connected graph with $\ell \le 4$ therefore satisfies Conjecture 133 of *Written on the Wall II*; the case that contains a triangle uses a computer-assisted finite certificate. The same arguments determine the maximum of ℓ over connected graphs with no 4-cycle subgraph and no induced path on six vertices, with a unique extremal graph for every order $n \ge 7$, classify the connected graphs with no 4-cycle subgraph of radius 4 and induced-path order 7, and determine, for every $n \ge 10$, the maximum number of edges in a connected triangle-free graph with no 4-cycle subgraph and no induced path on seven vertices, together with all extremal graphs. The conjecture remains open for $\ell > 4$.

1 | Introduction

All graphs are finite and simple. For a graph G with $n \ge 2$ vertices let $p(G)$ be the largest number of vertices of an induced path, $r(G)$ the radius, and

$$\ell(G) = \frac{1}{n} \sum_{v \in V(G)} \alpha\big(G[N(v)]\big),$$

where α is the independence number and $N(v)$ the open neighbourhood. Write $S(G)$ for the sum in the numerator, and write $c_{C_4}(G) = 1$ if G contains no 4-cycle as a subgraph and $c_{C_4}(G) = 0$ otherwise. A graph with no 4-cycle subgraph is called *strong C_4-free*.

Conjecture 1.1 (Graffiti/WOWII Conjecture 133)**.** *Every connected graph G on at least two vertices satisfies*

$$r(G) + \ell(G)^{c_{C_4}(G)} \le p(G).$$

The conjecture is number 133 in E. DeLaVina's list *Written on the Wall II* [5]. The formal statement used here is the one in the Formal Conjectures repository at commit 5faef0c4dc3d226f8f9a9fc8f581b6221283eb6e [7], the merge commit of pull request #5727 [8], which removed the floor function. That statement reads `(rad : ℝ) + 1 G ^ cC4 ≤ (path G : ℝ)`. We call this real-valued form REAL-133. If G contains a 4-cycle, a geodesic gives $p(G) \ge r(G) + 1$. Only strong C_4-free graphs remain, and for those the conjecture is equivalent to

$$p(G) - r(G) \ge \lceil \ell(G) \rceil. \tag{1.1}$$

Four arguments organise the theorems. The first is the two-layer analysis of an induced 6-cycle. Theorem 3.6 determines $M(n)$, the maximum of ℓ over connected strong C_4-free P_6-free graphs, and proves that the extremal graph is unique for every $n \geq 7$. The same theorem gives $S \leq 3n$ for every $n \geq 2$, with equality only for the Petersen graph P. The jump from order 10 to order 11 is the change from three occupied antipodal pairs to a single such pair. The second is the radius-4 classification. Theorem 4.3 shows that a connected strong C_4-free graph of radius 4 and induced-path order 7 is C_8 or the 6-cycle extension in the statement, with $S = 2n - b - m$, and in particular $\ell \leq 2$. The third passes from deletion loss to one graph: Lemma 5.7 identifies $H_{3,2}$ as the only connected strong C_4-free P_7-free graph that contains a triangle, meets $L(v) \geq 4$ at every vertex and has no two adjacent vertices of degree two, and Theorem 5.8 extends $S \leq 3n$, with equality only for P, from the triangle-free case of Theorem 5.5 to every connected strong C_4-free P_7-free graph. The fourth is Theorem 5.6, which determines the maximum number of edges, and all extremal graphs, over connected triangle-free strong C_4-free P_7-free graphs of every order $n \geq 10$. The cores come from Chen and Xu. What is new is the attachment of trees, the comparison of deficits, and the equality classification. The extremal family changes at orders 15 and 16.
Theorem 5.9 is REAL-133 for every connected graph with $\ell \leq 4$. The range $\ell > 4$ remains open, and no structural classification for radius $r \geq 5$ is proved here. Table 3 and Section 7 are the boundary.

Chen [3]: hypothesis and range	What changes	Extremal or uniqueness
Lemma 2.1. In a C_4-free graph every neighbourhood is a matching plus isolated vertices, and $\ell = (2e - 3T)/n$.	The identity is Lemma 2.1. His C_4-free means that the graph has no 4-cycle subgraph.	None. The local count is not the new content of Theorem 3.6.
Lemmas 7.18 and Corollary 7.20. On an induced 6-cycle in a C_4-free graph with no induced P_6, an outside vertex has 0 or 2 neighbours on the cycle, and if two they are antipodal. The class RES of Definition 7.2 is not a hypothesis.	The trace is kept.	Theorem 3.6 adds the second-layer rule, the function $M(n)$, and uniqueness for every $n \geq 7$.
Lemma 7.22, under RES $\wedge$ (C_6): radius 2, diameter 3, $\ell > 3$, condition (R3b), and no induced P_6. The lemma restricts the second layer.	The RES hypotheses are dropped; P_6-freeness is retained.	The second-layer rule holds for every connected strong C_4-free P_6-free graph.
Remark 7.24, p. 23. Five admissible frames, of orders 6, 7, 8, 9 and densities 2, 16/7, 5/2 and 8/3, with maximum $8/3 < 3$, from the frame enumeration in that radius-2, diameter-3, (R3b) analysis.	The frames are not an unrestricted classification. Orders 10 and $n \geq 11$ are absent.	Unique extremal graphs at those orders as well: P at order 10, and Γ_{n-7} for every $n \geq 11$.
Theorem 6.7. If the diameter is 2 and $\ell > 3$, then there is an induced path on at least six vertices. The radius is then 2, so the path order is at least $r+4$.	Diameter 2 is not assumed. The range proved here is $\ell \leq 4$.	For $r = 2$ and $3 < \ell \leq 4$, Theorem 5.9 gives $p \geq r + \ell$.
Display (6.1) and Theorem 7.1. A connected C_4-free graph with $\ell \geq 3$ and path order less than $r + 3$ cannot be a minimum counterexample of radius at most 3.	Chen's target on this band is the integer bound $p \geq r + 3$. His conjecture is the floor form.	For $r = 3$ and $3 < \ell \leq 4$, Theorem 5.9 asks for $p \geq r + \ell$ and uses $S \leq 3n$ from Theorem 5.8.

TABLE 1 | Comparison with Chen [3], Zenodo https://doi.org/10.5281/zenodo.22054835. The rows are the statements used in the text. His Conjecture 133 is the floor form $p \geq r + \lfloor \ell \rfloor^{cc_4}$.

Lemma 2.9 is the emptiness list from Karthick and Maffray used in the induced 5-cycle branch; their colouring bound is a different statement. Brandstädt and Hoàng [1] describe (P_6, C_4)-free graphs by clique cutsets, with C_4 an induced cycle. A nontrivial blow-up of the Petersen graph, and a join of the Petersen graph with a nonempty clique, each contain a 4-cycle subgraph, so neither enters the class of Theorem 3.6, which does include graphs with clique cutsets. The cores of Theorems 5.5 and 5.6 are Lemma 5.1, from Chen and Xu. Lemma 2.8 is the case $k = 6$ of Camby and Schaudt. It is used in Theorem 4.3 and in the case $r = 3$ of Theorem 5.9.
Section 3 proves Theorem 3.6. Section 4 proves Theorem 4.3. Section 5 proves Theorems 5.5, 5.6 and 5.8, and Theorem 5.9. Section 6 is the reduction for $\ell > 4$. The supplement contains the rooted bounds, the endpoint obstructions, the high-radius constructions and the certificate. Which of those arguments is a hypothesis is recorded only in the dependency

note at the start of the supplement. The certificate for Lemma 5.7 is the checker `certificates/gen3/verify_gen3.py` at commit `dd555d305b4a69076cafb845087f00b4144f07cc`. From the repository root the command is `python3 certificates/gen3/verify_gen3.py`. Appendix A of the supplement, Section A, records an independent execution.

2 | Preliminaries

For $v \in V(G)$ let $d(v)$ be its degree and $t(v)$ the number of triangles containing v. Write $\lambda(v) = \lambda_G(v) = \alpha(G[N(v)])$, $S(G) = \sum_v \lambda(v)$, $e(G)$ for the number of edges and $T(G)$ for the number of triangles.

Lemma 2.1. *Let G be strong C_4-free. Then:*

1. *two distinct vertices have at most one common neighbour;*
2. *for every v, $G[N(v)]$ is a matching together with isolated vertices, so $\lambda(v)$ equals the number of components of $G[N(v)]$ and $\lambda(v) = d(v) - t(v)$;*
3. *$S(G) = 2e(G) - 3T(G)$.*

Proof. (1) Two common neighbours w, w' of x, y give the 4-cycle $xwyw'$. (2) A vertex of $N(v)$ with two neighbours in $N(v)$ would have two common neighbours with v. So the components are single vertices and edges, each edge being a triangle at v. (3) Sum $d(v) - t(v)$ over v. □

The components of $G[N(v)]$ are the *units* of v. For a vertex set U, $G - U$ is the subgraph induced by $V \setminus U$, and $D_G(U) = S(G) - S(G - U)$ is the *deletion loss* of U. We write $L(v) = D_G(\{v\})$.

Lemma 2.2. *Let G be strong C_4-free and $\emptyset \neq U \subseteq V(G)$. Then*

$$D_G(U) = \sum_{u \in U} \lambda(u) + \sum_{x \notin U} \#\{\text{units of } x \text{ contained in } U\}.$$

In particular $L(v) = 2\lambda(v) - t(v) = 2d(v) - 3t(v)$.

Proof. For $x \notin U$, deleting U removes exactly the units of x that lie inside U; a unit that is an edge with one end in U survives as a vertex. For $U = \{v\}$: the $2t(v)$ neighbours of v in triangles with v keep their value, and each of the other $d(v) - 2t(v)$ neighbours loses one. The total is $\lambda(v) + d(v) - 2t(v) = 2\lambda(v) - t(v)$. □

Lemma 2.3. *Let v be a cut vertex of G. Split the components of $G - v$ into two nonempty groups, and let G_1, G_2 be the subgraphs induced by the two groups together with v. Then $S(G) = S(G_1) + S(G_2)$.*

Proof. $G[N(v)]$ has no edges between the groups, so $\lambda_G(v) = \lambda_{G_1}(v) + \lambda_{G_2}(v)$. Every other vertex keeps its neighbourhood. □

We use three published results.

Theorem 2.4 (Erdős, Saks and Sós [6])**.** *Every connected graph G has an induced path on at least $2r(G) - 1$ vertices.*

Theorem 2.5 (Camby and Schaudt [2, Theorem 5])**.** *Let $k \geq 4$ and let G be a connected graph with no induced path on k vertices. Then G has a connected dominating set X such that $G[X]$ has no induced path on $k-2$ vertices or $G[X]$ is isomorphic to the cycle C_k.*

Theorem 2.6 (Seinsche [10])**.** *If a graph on at least two vertices has no induced path on four vertices, then it or its complement is disconnected.*

Theorem 2.5 is the implication (i)⇒(ii) of [2, Theorem 5] applied with $H = G$. We write P_k for the path on k vertices, and "G is P_k-free" means that G has no induced P_k, that is, $p(G) \leq k - 1$. A connected graph of radius r has at least $2r$ vertices.

Lemma 2.7. *Let H be a connected strong C_4-free P_4-free graph on at least two vertices. Then H has a vertex adjacent to all other vertices.*

Proof. By Theorem 2.6, H is the join of two nonempty vertex sets A and B. If $|A|, |B| \geq 2$, then $a, a' \in A$ and $b, b' \in B$ span the 4-cycle $aba'b'$. So one side is a single vertex, and it is adjacent to everything. □

Lemma 2.8. *Let G be a connected graph with no induced path on six vertices. Then G has a connected dominating set X such that $G[X]$ has no induced path on four vertices, or else $G[X]$ is isomorphic to C_6.*

This is Theorem 2.5 at $k = 6$. Lemma 3.5 uses the resulting dominating set.

Lemma 2.9 (Karthick and Maffray [9, Theorem 3.2(d)]). *Let G be a graph with no induced P_6, no induced C_4 and no induced C_6, and let $C = c_0c_1c_2c_3c_4c_0$ be an induced 5-cycle, indices modulo 5. For each i let W_i be the set of vertices outside C whose neighbourhood on C is exactly $\{c_i\}$, and let $X_{i,i+1}$ be the set of vertices outside C whose neighbourhood on C is exactly $\{c_i, c_{i+1}\}$. Then, for every i, at least one of $X_{i,i+1}$ and $X_{i+1,i+2}$ is empty, at least one of W_i and W_{i+2} is empty, and at least one of $X_{i,i+1}$ and $W_{i-1} \cup W_{i+2}$ is empty.*

In [9, Theorem 3.2] the cycle is labelled $v_1, \dots, v_5$ and C_4 denotes an induced cycle. The translation $v_{i+1} = c_i$ carries their sets onto W_i and $X_{i,i+1}$ as defined here, and the three emptiness assertions are the whole of their part (d). A strong C_4-free graph has no induced C_4, so it meets the induced-cycle hypothesis whenever it is P_6-free. Case 2 of Theorem 3.6 uses the first two assertions. The third is included so that the quoted part is complete.

Lemma 2.10. *Let G be a connected triangle-free strong C_4-free P_7-free graph with no cut vertex. Then G has no clique cutset.*

Proof. The graph is triangle-free, so every clique has order at most two, and there is no clique cutset of order one. Suppose $\{u, v\}$ is a clique cutset. Then uv is an edge. Neither vertex is a cut vertex, so every component of $G - \{u, v\}$ is adjacent to both u and v, and there are at least two such components. A shortest u–v path through one component has at least four edges: one internal vertex would form a triangle with uv, and two internal vertices would form a 4-cycle with uv. On each of two such paths, let x, y, z be the three vertices immediately preceding v, in that order from the u-side, and write x', y', z' for the same choice in the second component. The path x, y, z, v, z', y', x' is an induced P_7. Each side is a subpath of a shortest path, so it has no chord, and there is no edge between $\{x, y, z\}$ and $\{x', y', z'\}$ because those sets lie in different components of $G - \{u, v\}$. This contradicts $p(G) \le 6$. □

3 | Small-path mass bounds

Lemma 3.1. *If G is connected and strong C_4-free with $r(G) = 1$, then $S(G) = 2(n-1) - m$, where m is the number of edges of $G[N(u)]$ for a dominating vertex u. Hence $\ell(G) < 2$, and (1.1) holds.*

Proof. Let $v \neq u$. By Lemma 2.1(1), v has at most one neighbour other than u, and such a neighbour lies in $N(u)$. So $G[N(v)]$ is a clique and $\lambda(v) = 1$; also $\lambda(u) = (n-1) - m$. Hence $S = 2(n-1) - m < 2n$ and $\lceil \ell \rceil \le 2$. If G is complete then $G \in \{K_2, K_3\}$, $\ell = 1$ and $p = 2 = r + 1$. Otherwise two nonadjacent vertices of $N(u)$ together with u form an induced P_3, so $p \ge 3 \ge r + \lceil \ell \rceil$. □

Lemma 3.2. *If G is connected and strong C_4-free with $r(G) \ge 2$, then $p(G) \ge 4$.*

Proof. If G were P_4-free, Lemma 2.7 would give a dominating vertex, so $r(G) \le 1$. □

Lemma 3.3. *If G is connected, strong C_4-free and P_5-free, then $S(G) \le 2n - h$, where h is the number of vertices v with $\lambda(v) \ge 3$. In particular $S(G) \le 2n$.*

Proof. Call v *simplicial* if $\lambda(v) = 1$ and *high* if $\lambda(v) \ge 3$. For $x \in N(v)$ put $X_x = N(x) \setminus N[v]$; then $N(x) \subseteq \{v\} \cup \{\bar{x}\} \cup X_x$, where $\bar{x}$ is the partner of x if x lies in an edge unit of v.
Key fact. Let x, y lie in different units of v. A common vertex of X_x and X_y would give the 4-cycle $vxwy$, so $X_x \cap X_y = \emptyset$. If $x' \in X_x$ and $y' \in X_y$ are nonadjacent, then $x'xvyy'$ is an induced P_5: $x \nsim y$ because they lie in different units, $x \nsim y'$ and $y \nsim x'$ by disjointness, and v is adjacent to neither x' nor y'. So X_x is complete to X_y. If both are nonempty and $|X_x| \ge 2$, a vertex of X_y has two neighbours in X_x, which are also neighbours of x, a contradiction to Lemma 2.1(1). Hence if both are nonempty, then $|X_x| = |X_y| = 1$.
Main step. For each high v we find $\lambda(v) - 1$ simplicial vertices whose only high neighbour is v, one in each of $\lambda(v) - 1$ distinct units of v.
Case 1: some $u \in N(v)$ has $|X_u| \ge 2$. By the key fact $X_y = \emptyset$ for every y outside the unit of u. Then $N(y) \subseteq \{v, \bar{y}\}$, which is a clique, so y is simplicial; so is $\bar{y}$. Hence v is the only high neighbour of y, and the $\lambda(v) - 1$ units other than that of u supply the vertices.
Case 2: $|X_x| \le 1$ for all $x \in N(v)$. Then $\lambda(x) \le 2$ for $x \in N(v)$. Call a unit *active* if one of its vertices x has $X_x \neq \emptyset$. Suppose two units are active, with $y' \in X_y$ and $z' \in X_z$, and let x lie in a third unit. By the key fact $y' \sim z'$, and $xvyy'z'$ is an induced P_5: $x \nsim y$ (different units), $x \nsim y', z'$ (disjointness), $v \nsim y', z'$, and $y \nsim z'$ (disjointness). As $\lambda(v) \ge 3$, a third unit exists, so at most one unit is active. Every vertex of an inactive unit is simplicial and its only high neighbour is v. This gives at least $\lambda(v) - 1$ units.

Counting. The chosen vertices for distinct high vertices are distinct, since each has a unique high neighbour. So $\sum_{v \text{ high}}(\lambda(v)-1) \le \#\{\text{simplicial vertices}\}$, and

$$S = \sum_{v} \lambda(v) \le 2n - \#\{\text{simplicial}\} + \sum_{v \text{ high}} (\lambda(v)-2) \le 2n - h. \qquad \square$$

Lemma 3.4. *Let G be connected, strong C_4-free and P_6-free with a dominating induced cycle $C = v_0v_1\cdots v_5$, and let $W = V(G)\setminus V(C)$. Then every $w \in W$ is adjacent to exactly one antipodal pair $\{v_i, v_{i+3}\}$ and to no other vertex of C, distinct vertices of W use distinct pairs, and W is independent. Hence $|W| \le 3$, $S(G) = 12 + 4|W| < 3n$, and $r(G) \le 2$ if $W \ne \emptyset$.*

Proof. Indices are modulo 6. Let $w \in W$; it has a neighbour on C. If w has exactly one neighbour v_i, then $wv_iv_{i+1}v_{i+2}v_{i+3}v_{i+4}$ is an induced P_6. If $N(w)\cap C = \{v_i, v_{i+1}\}$, then $wv_{i+1}v_{i+2}v_{i+3}v_{i+4}v_{i+5}$ is an induced P_6. Two neighbours at distance 2 on C give a 4-cycle. A set of at least three vertices of C_6 contains two vertices at distance 2. So $N(w)\cap C$ is an antipodal pair. Two vertices of W with the same pair give two common neighbours. If $w \sim w'$ with distinct pairs, some $v_i \in N(w)$ and $v_j \in N(w')$ are adjacent on C, and wv_iv_jw' is a 4-cycle. There are no triangles, $e(G) = 6 + 2|W|$, and Lemma 2.1(3) gives $S = 12 + 4|W|$. Finally, if w uses $\{v_i, v_{i+3}\}$, then every vertex of C is within distance 2 of v_i (the vertex v_{i+3} through w), and each $w' \in W$ has a neighbour in $\{v_{i-1}, v_i, v_{i+1}\}$, because each of the three antipodal pairs meets this set. So v_i has eccentricity 2. $\square$

The same trace analysis applies to any vertex adjacent to an induced 6-cycle in a connected strong C_4-free P_6-free graph, whether or not the cycle dominates. Theorem 3.6 uses that form. Karthick and Maffray [9, Theorem 3.3] classify the external neighbourhood of an induced C_6 in a (P_6, C_4)-free graph. When the graph has no 4-cycle subgraph, only the empty trace and the antipodal-pair traces remain. The new content here is the second-layer constraint, the resulting order bound, the exact value of $M(n)$, and uniqueness.

Lemma 3.5. *Let G be connected, strong C_4-free and P_6-free. Then either G has a vertex of eccentricity at most 2, or $G = C_6$.*

Proof. Lemma 2.8 supplies a connected dominating set X. If $G[X] \cong C_6$, Lemma 3.4 gives $G = C_6$ or a vertex of eccentricity 2. Otherwise $G[X]$ is P_4-free. If $|X| = 1$, the vertex in X dominates G. If $|X| \ge 2$, Lemma 2.7 gives $u \in X$ adjacent to all of $X \setminus \{u\}$, and since X dominates G, every vertex is within distance 2 of u. $\square$

Theorem 3.6. *Let $M(n)$ be the maximum of $\ell(G)$ over connected strong C_4-free P_6-free graphs of order $n \ge 7$. Then*

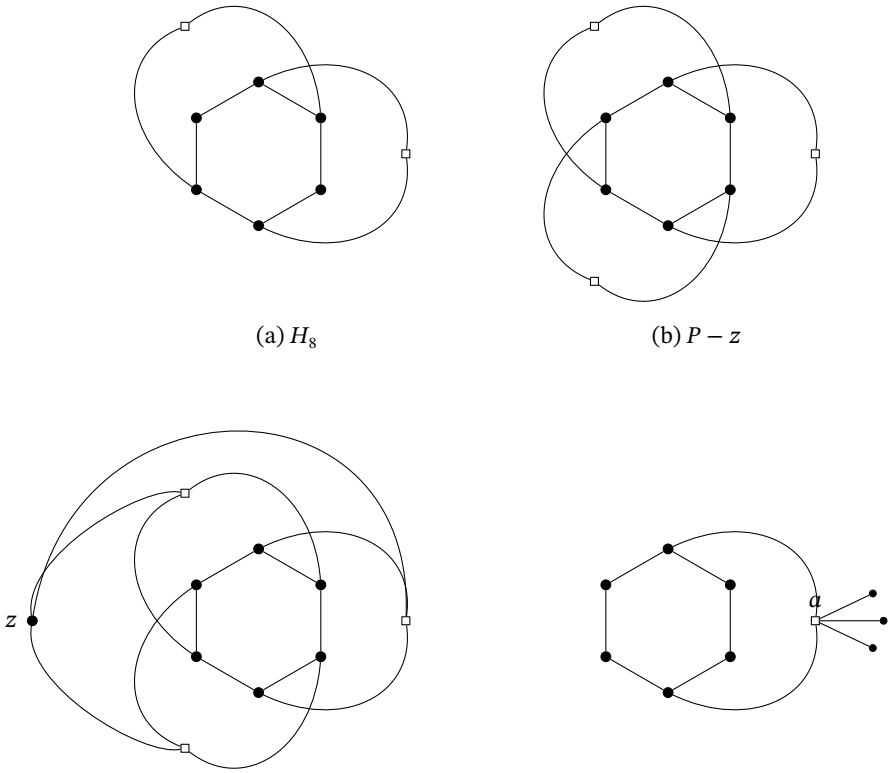
$$M(n) = \begin{cases} 16/7 & \text{if } n = 7, \\ 5/2 & \text{if } n = 8, \\ 8/3 & \text{if } n = 9, \\ 3 & \text{if } n = 10, \\ 2 + 2/n & \text{if } n \ge 11. \end{cases}$$

For every $n \ge 7$ the extremal graph is unique up to isomorphism. For $n = 10$ it is the Petersen graph P. For $n = 9$ it is $P - v$. For $n = 8$ it is the graph H_8 of Figure 1. For $n = 7$ and for every $n \ge 11$ it is the graph Γ_{n-7} of that figure: a 6-cycle, a vertex a adjacent exactly to one antipodal pair, and $n - 7$ leaves adjacent only to a. In particular, every connected strong C_4-free P_6-free graph G of order $n \ge 2$ satisfies $S(G) \le 3n$, with equality if and only if G is the Petersen graph. At orders 4 and 6 the maximum is attained by two nonisomorphic graphs.

Proof. Throughout, $S = 2e - 3T$ by Lemma 2.1. A chordless cycle of length at least seven contains an induced P_6. Thus the only possible holes have lengths five and six.

Case 1: an induced six-cycle. Label its vertices $c_0, \dots, c_5$, indices modulo 6, and write A for the set of vertices at distance one from the cycle. The trace argument in the proof of Lemma 3.4 shows that each $a \in A$ meets the cycle in one antipodal pair. Distinct vertices of A use distinct pairs, and A is an independent set: an edge of A joins distinct pairs, and some contact of one end is adjacent on the cycle to some contact of the other, giving a 4-cycle. Hence $k = |A| \le 3$. Figure 1 shows the three extremal graphs of this branch, and Figure 2 shows the two distance layers used in the count.

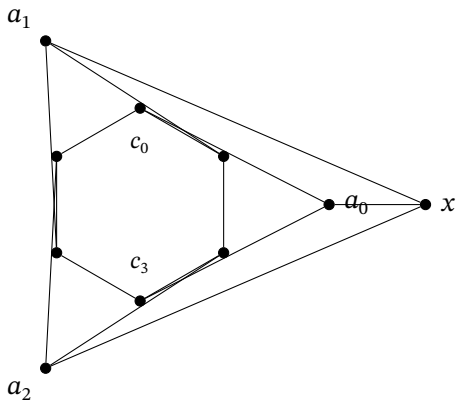
Every vertex x at distance two from the cycle is adjacent to every vertex of A. Such an x has at least one neighbour in A. If $x \sim a$ and $x \nsim b$ for $a, b \in A$, choose contacts so that $c_i \in N(a)$ and $c_{i+2} \in N(b)$. This is possible: the pairs are distinct, so some contact of b lies at cycle distance two from some contact of a, and we orient the cycle from that contact of a toward that contact of b. The path $x, a, c_i, c_{i+1}, c_{i+2}, b$ is then an induced P_6. Indeed x meets the cycle in nothing, a meets the cycle only in its antipodal pair (neither c_{i+1} nor c_{i+2}), b meets the cycle only in its antipodal pair (so $b \sim c_{i+2}$ and $b \nsim c_i, c_{i+1}$), and $a \nsim b$.



(a) H_8 (b) $P - z$

(c) P (d) Γ_h

FIGURE 1 | Extremal graphs for $M(n)$. The six central filled vertices form the cycle, and each open square meets one antipodal pair. (a) H_8, order 8, $\ell = 5/2$. (b) $P - z$, order 9, $\ell = 8/3$. (c) The Petersen graph, order 10, $\ell = 3$. These three graphs are triangle-free, so $S = 2e$. (d) Γ_h: any number of leaves at the square a, including none. The case $h = 0$ is the unique graph of order 7 with $\ell = 16/7$. For every $n \geq 11$ the extremal graph is Γ_{n-7}, with $S = 2n+2$.



first layer a_i second layer x

FIGURE 2 | Distance layers of an induced 6-cycle in Case 1 of Theorem 3.6. Each a_i meets one antipodal pair, the pairs are distinct, and the first layer is independent, so $k \leq 3$. A vertex x of the second layer meets every vertex of the first layer and none of the cycle. If $k \in \{2, 3\}$ that layer has size at most one, and the order is at most 10. There is no third layer.

There is no vertex at distance three or more from the cycle. If there were, the cycle would have a vertex z at distance exactly three. A shortest z–cycle path is then a four-vertex geodesic z, x, a, c_i, with x at distance two and a at distance one. Extending that geodesic by c_{i+1}, c_{i+2} gives an induced P_6: the vertex z has no neighbour in A or on the cycle, and x has no neighbour on the cycle.

Let h be the number of vertices at distance two. If $k = 0$, then A is empty, so the cycle has no neighbour and $G = C_6$. If $k = 1$, every distance-two vertex is adjacent to the unique vertex $a \in A$. The distance-two vertices induce a matching together

with isolated vertices: a vertex of that layer with two neighbours in the layer would have two common neighbours with a. Writing m for the number of edges of the matching gives $n = 7+h$, $e = 8+h+m$ and $T = m$, so $S = 16+2h-m = 2n+2-m$. Thus $S \leq 2n + 2$, with equality if and only if $m = 0$. Equality means that G is the graph Γ_h formed by the 6-cycle, the vertex a on one antipodal pair, and h leaves at a.

The graph Γ_h lies in the class, for every $h \geq 0$. A leaf has degree one. Every other cycle through an outside vertex uses both edges from a to the cycle, and each of the two a-paths along the cycle has length three, so every such cycle has length five. There is no 4-cycle. There is no induced P_6. A leaf, having degree one, occupies an end of any path that contains it, and that path meets a next. Two leaves therefore determine a path on only three vertices. An induced path that contains one leaf and five further vertices would restrict to an induced path of order five in Γ_0 that starts at a. No such path exists: the two neighbours of a on the cycle are an antipodal pair, the path $a, c_i, c_{i+1}, c_{i+2}, c_{i+3}$ has the chord ac_{i+3}, and every shorter continuation from a through one neighbour stops at the vertex before the antipode, because the next cycle vertex is adjacent to a. An induced path of Γ_0 that avoids a lies on the 6-cycle and has order at most five. An induced path of Γ_0 that contains a in its interior has c_i and c_{i+3} as the two neighbours of a on the path. Each side can then take at most one further cycle vertex: the next vertex on either arc is adjacent to the antipode already used. The resulting path has order at most five. Thus Γ_h is strong C_4-free and P_6-free, and it attains $S = 2n + 2$.

If $k \in \{2, 3\}$, then $h \leq 1$: two vertices of the second layer would both be adjacent to two vertices of A. There are no triangles. The second-layer vertex, if it exists, is adjacent to every vertex of A and to nothing on the cycle, A is independent, and each vertex of A contributes two edges to the cycle. Thus $n = 6 + k + h$ and $S = 2(6 + 2k + kh)$. The four possibilities are

(k, h)	$(2, 0)$	$(2, 1)$	$(3, 0)$	$(3, 1)$
S/n	$5/2$	$8/3$	$8/3$	3

The four pairs are the graphs of Figure 1, with $(3, 0)$ and $(2, 1)$ both equal to $P - z$. Write a_i for the vertex adjacent to the antipodal pair $\{c_i, c_{i+3}\}$. The pair $(k, h) = (3, 1)$ is the graph in panel (c): the three vertices a_0, a_1, a_2 and a vertex z adjacent to all three. It is cubic of order 10. It is triangle-free, and the trace rules give no 4-cycle, so its girth is at least 5. Its diameter is 2. Indeed every pair of cycle vertices is at distance at most 2 along the cycle or through the corresponding a_i; the vertex z meets every a_i; and a non-neighbour of a_i on the cycle is adjacent to c_i or to c_{i+3}. Suppose $v_0 \cdots v_5$ were an induced path. Diameter 2 gives v_0 a common neighbour with each of v_3, v_4, v_5. Those neighbours lie off the path, because the only neighbour of v_0 on the path is v_1 and the path is induced. They are distinct: a vertex adjacent to two of v_3, v_4, v_5 would form a triangle with a path edge, or a 4-cycle with the length-2 subpath $v_3v_4v_5$. This is impossible: v_0 has degree 3, so it has only two neighbours off the path. Thus the graph is P_6-free. It is the Petersen graph P, and $\ell(P) = 3$.

Deleting z leaves $(k, h) = (3, 0)$, the graph $P-z$ in panel (b). The pair $(2, 1)$ is isomorphic to the same graph. In the labelling with $A = \{a_0, a_1\}$ and second-layer vertex z, the cycle $c_0, a_0, c_3, c_4, a_1, c_1$ has its three antipodal pairs served by c_5, z and c_2, and no vertex remains. Both graphs therefore have $\ell = 8/3$, and both are isomorphic to $P - v$. Induced subgraphs of P are strong C_4-free and P_6-free. The pair $(2, 0)$ is the graph H_8 in panel (a), with $\ell = 5/2$. The three choices of the unused antipodal pair differ by a rotation, so this graph is unique up to isomorphism. Its order is 8, so it is neither P nor $P - v$. For $k \in \{2, 3\}$ one has $n = 6 + k + h \leq 10$.

Case 2: an induced five-cycle and no induced six-cycle. Write $C = c_0 \cdots c_4c_0$. Every vertex at distance one from C has a singleton or consecutive-pair trace: any other pair of contacts lies at distance two on C and forms a 4-cycle. If a vertex x were at distance two, with a neighbour a in the first layer, the path x, a followed by four consecutive cycle vertices, beginning at one contact of a and omitting the second contact if a has one, would be an induced P_6. Thus C dominates G. Write h for the number of vertices with singleton trace and j for the number with a consecutive-pair trace. At most one vertex carries each of the five pair traces, since two would have two common neighbours, so $j \leq 5$. The edges outside C are of the following kinds.

- Two singleton vertices with different supports are nonadjacent. Neighbouring supports give a 4-cycle. Supports at distance two give an induced 6-cycle along the longer arc of C. Edges within one singleton class form a matching, because two neighbours in that class would be two common neighbours of the supporting cycle vertex. Let m be the number of these edges.
- No two pair vertices are adjacent. From their two distinct support edges one can choose adjacent cycle vertices, one from each edge, and those two vertices together with the pair vertices form a 4-cycle.
- A singleton supported at c_i meets only a pair vertex supported on the opposite edge $\{c_{i+2}, c_{i+3}\}$. Every other pair edge contains a neighbour of c_i and yields a 4-cycle. That pair vertex meets at most one vertex of the singleton class, again by the common-neighbour bound. Let b be the number of singleton-to-pair edges, so $b \leq \min(h, j)$.

Consequently $n = 5+h+j$ and $e = 5+h+2j+m+b$. The triangles are the j pair triangles and the m matching triangles, so $T = j + m$ and

$$S = 2e - 3T = 2n + 2b - j - m.$$

The integer b is at most 1. Suppose two singleton–pair edges are present. They use two distinct singleton supports. The opposite edge of c_i is $\{c_{i+2}, c_{i+3}\}$. As i varies, these five edges exhaust C, and each pair class contains at most one vertex. The graph is P_6-free and strong C_4-free, and the present case assumes that it has no induced C_6, so Lemma 2.9 applies. Under that lemma the set W_i is the singleton class at c_i, and the set $X_{i,i+1}$ is the consecutive-pair class on the edge $\{c_i, c_{i+1}\}$. If the two singleton supports lie at distance 2 on C, they are W_i and W_{i+2} for some i, and both are nonempty. Lemma 2.9 states that at least one of those two sets is empty. If the supports lie at distance 1, take them to be c_i and c_{i+1}. The opposite edge of c_i is $\{c_{i+2}, c_{i+3}\}$ and the opposite edge of c_{i+1} is $\{c_{i+3}, c_{i+4}\}$. Both pair classes are nonempty, one for each of the two edges, and they are $X_{i+2,i+3}$ and $X_{i+3,i+4}$. Lemma 2.9 states that at least one of $X_{j,j+1}$ and $X_{j+1,j+2}$ is empty for every j; the index $j = i + 2$ is this pair of classes. Either alternative contradicts the existence of both edges, so $b \le 1$. The third emptiness assertion of Lemma 2.9 is not required for the count.

If $b = 0$, then $S = 2n - j - m \le 2n$. If $b = 1$, then $j \ge 1$ because $b \le \min(h, j)$, so $S = 2n + 2 - j - m \le 2n + 1$. In this case $S \le 2n + 1$.

Case 3: no holes. The graph is chordal, and every nonempty induced subgraph has a simplicial vertex. The neighbourhood of a simplicial vertex is a clique. The degree is at most two: four vertices of a clique contain a 4-cycle. A simplicial vertex of a connected chordal graph on at least two vertices is not a cut vertex, and the graph left by deleting it is chordal and strong C_4-free. In the graph present at the deletion, the loss is $L(v) = 2d(v) - 3t(v)$. Degree one gives loss 2. Degree two, with the two neighbours adjacent, gives one triangle and loss 1. Deleting vertices until a single vertex remains, one has $S \le 2(n-1) = 2n - 2$.

The three cases give the values of $M(n)$. A graph of order $n \ge 11$ cannot realise $k \in \{2, 3\}$ in Case 1, and $k = 0$ forces $G = C_6$. The only remaining graphs in Case 1 are the graphs Γ_h and the graphs with the same first layer and $m \ge 1$. The latter have $S \le 2n + 1$. Cases 2 and 3 give $S \le 2n + 1$ and $S \le 2n - 2$. Thus $M(n) = 2 + 2/n$, attained only by Γ_{n-7}.

The graphs with $S > 2n + 2$ are as follows. At order $n = 10$ the only graph in the list is P, with $S = 30$. At order $n = 9$ the only such graph is $P - v$, with $S = 24$, whether it arises as $(k, h) = (3, 0)$ or as $(k, h) = (2, 1)$. At order $n = 8$ the only such graph is H_8, with $S = 20$. At order $n = 7$ every graph in the list satisfies $S \le 2n + 2$. One has $h = 0$ in the equality case of $k = 1$, so the unique graph with $S = 2n + 2$ is Γ_0, while Cases 2 and 3 give at most 15 and 12. Since $8/3 < 3$ and $2 + 2/n < 3$ for $n \ge 11$, the bound $S \le 3n$ holds for every order $n \ge 2$, with equality only for P.

The case division does not use $n \ge 7$. At order 4 there is no cycle of length five or six, so only the chordal bound applies and $S \le 6$. Equality holds for a tree, and the only trees on four vertices are P_4 and $K_{1,3}$. At order 6 the induced 6-cycle branch contributes only C_6, with $S = 12$. In the induced 5-cycle branch one has $h + j = 1$, hence $b = 0$ and $S = 12 - j - m \le 12$, with equality precisely when $j = m = 0$ and $h = 1$, the graph C_5 with one leaf. The chordal branch contributes at most 10. Thus the maximum is 12, attained by exactly two graphs. □

4 | Radius four

Lemma 4.1. *Let G be connected, strong C_4-free and P_8-free. If an induced 8-cycle dominates G and $r(G) \ge 4$, then $G = C_8$.*

Proof. Write $C = c_0c_1 \cdots c_7c_0$, with indices modulo 8. For an outside vertex y the trace $A_y = N(y) \cap V(C)$ is nonempty. The trace A_y is not contained in any three consecutive vertices of C. Suppose $A_y \subseteq \{c_i, c_{i+1}, c_{i+2}\}$. The ends c_i and c_{i+2} are not both in A_y, since y, c_i, c_{i+1}, c_{i+2} would be a 4-cycle. Thus A_y is a singleton or a consecutive pair. A singleton $\{c_i\}$ gives the induced path

$$y, c_i, c_{i+1}, c_{i+2}, c_{i+3}, c_{i+4}, c_{i+5}, c_{i+6}.$$

A consecutive pair $\{c_i, c_{i+1}\}$ gives the induced path

$$y, c_i, c_{i-1}, c_{i-2}, c_{i-3}, c_{i-4}, c_{i-5}, c_{i-6},$$

in which the omitted contact c_{i+1} does not lie on the path. Either path is an induced P_8.

Fix a cycle vertex, say c_0. The vertices at cycle distance greater than two from c_0 are the three consecutive vertices c_3, c_4, c_5. The trace of any outside vertex meets the complementary set $\{c_7, c_0, c_1, c_2, c_6\}$, so every outside vertex is within distance three of c_0. The same holds with c_0 replaced by any vertex of C.

Suppose w lies outside C, and write $A = N(w) \cap V(C)$. No two vertices of A are at cycle distance two, for they would form a 4-cycle with w. A set of vertices of C in which every two distinct vertices are consecutive has size at most two, and every set of size at most two that is a singleton or a consecutive pair is contained in a triple of consecutive vertices. Both

possibilities have been excluded. Therefore A contains two vertices at cycle distance three or four. Rotating, take them to be c_0 and c_d with $d \in \{3, 4\}$.
The antipode of c_0 is c_4. If $d = 4$, the path $c_0 w c_4$ has length two. If $d = 3$, then $c_3 \sim c_4$, so $c_0 w c_3 c_4$ has length three. Every other vertex of C is already at cycle distance at most three from c_0. Every outside vertex is within distance three of c_0. Thus the eccentricity of c_0 is at most three, contradicting $r(G) \geq 4$.
Hence C has no vertex outside it, and $G = C_8$. □

Remark 4.2. The proof of Lemma 4.1 is the trace argument above. An independent count, with the labelled cycle fixed, is the script `paper/scripts/c8_configurations.py`. A trace is admissible when the graph formed by the cycle and one vertex of that trace has no 4-cycle subgraph and no induced P_8. Configurations are built by adding admissible traces in increasing bitmask order, joining the new vertex to an arbitrary subset of the outside vertices already chosen, and discarding any graph that acquires a 4-cycle subgraph or an induced P_8. The script finds 32 admissible nonempty traces and 790 configurations, including the bare cycle. By order $8, \ldots, 13$ the counts are $1, 32, 262, 384, 107, 4$. It finds 73 isomorphism classes, and only the bare cycle has radius at least four.

Theorem 4.3. *A connected graph G is strong C_4-free and satisfies $r(G) = 4$ and $p(G) = 7$ if and only if either $G \cong C_8$, or G is obtained from a 6-cycle $C = v_0 v_1 \cdots v_5$ by the following construction and by no further vertices or edges. For each i there is a nonempty set C_i of vertices outside C. Every vertex of C_i is adjacent to v_i and to no other vertex of C. There is no edge between C_i and C_j for $i \neq j$, and $G[C_i]$ is a matching together with isolated vertices. For each edge of C there is at most one further vertex, adjacent precisely to the two ends of that edge.*

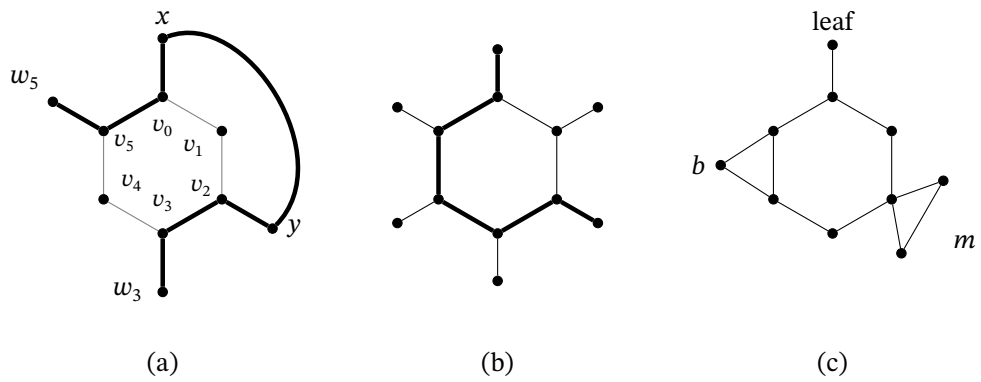


FIGURE 3 | The radius-four case of Theorem 4.3. (a) The induced P_8 is $w_5, v_5, v_0, x, y, v_2, v_3, w_3$, with singleton traces $\{v_0\}$ and $\{v_2\}$ at x and y, and with w_5, w_3 the distance-four witnesses. (b) One leaf at each cycle vertex gives $S = 2n$. (c) Only local attachments are shown; the remaining nonempty attachment classes are omitted. An isolated leaf does not contribute to $b + m$. A matching edge in some C_i, and a vertex glued to a cycle edge, each increase $2n - S = b + m$ by 1.

Lemma 4.4. *Let G be obtained from a 6-cycle C by the construction of Theorem 4.3. Let P be an induced path in G of order at least three. Every vertex of P that does not lie on C is an endpoint of P. In particular P contains at most two vertices outside C, and any such vertex is an endpoint of P.*

Proof. Let z be a vertex of P outside C. If z is an isolated vertex of $G[C_i]$ for some i, then z has degree one, so z is an endpoint of every path that contains it.
If z lies on a matching edge of $G[C_i]$, the neighbourhood of z is $\{v_i, z'\}$, where z' is the other end of that edge, and this neighbourhood is a clique. Suppose z were internal on P. An internal vertex has two neighbours on P. Those neighbours are then v_i and z', which are adjacent, and the path uses the route $v_i z z'$ or $z' z v_i$. The edge $v_i z'$ joins the two neighbours of z and skips z, so it is a chord of P. An induced path has no chord. Thus z is an endpoint of P. The same chord shows that P does not contain z, z' and v_i together.
If z was added on a cycle edge, the neighbourhood of z is that edge, again a clique. The same chord argument shows that z is an endpoint of P, and that P does not contain both ends of the cycle edge together with z.
A path has two endpoints. Every vertex of P outside C is one of them, so there are at most two such vertices. □

Proof of Theorem 4.3. First suppose that G is strong C_4-free, with $r(G) = 4$ and $p(G) = 7$. Apply Theorem 2.5 with $k = 8$. The hypotheses hold: $k \geq 4$, and G is connected and has no induced path on eight vertices because $p(G) = 7$. The conclusion is a connected dominating set X such that $G[X]$ is P_6-free or $G[X] \cong C_8$.

If $G[X] \cong C_8$, Lemma 4.1 gives $G = C_8$.
If $G[X]$ is P_6-free, apply Lemma 3.5 inside $G[X]$. A vertex of eccentricity at most two in $G[X]$ has eccentricity at most three in G, because X dominates G, contradicting $r(G) = 4$. Therefore $G[X] = C_6$. Let $C = v_0 \cdots v_5$ be this dominating induced 6-cycle, indices modulo 6, and let $W = V(G) \setminus V(C)$.
Traces. As in the proof of Lemma 3.4, strong C_4-freeness allows only three kinds of trace $N(w) \cap C$: a single vertex, a consecutive pair $\{v_i, v_{i+1}\}$, or an antipodal pair. If some w has trace $\{v_i, v_{i+3}\}$, then v_i is within distance two of every vertex of C, hence within distance three of every vertex of G, a contradiction. So there are no antipodal traces. Let C_i be the set of vertices with trace $\{v_i\}$.
Witnesses. Fix i. Since $r(G) = 4$, some vertex is at distance at least four from v_{i+3}. Every vertex of C is within distance three of v_{i+3}, and a vertex $w \in W$ is at distance at most $1 + \min\{d_C(v_{i+3}, x) : x \in N(w) \cap C\}$ from v_{i+3}, which is at most three unless the trace of w is $\{v_i\}$. So there is $w_i \in C_i$ at distance four from v_{i+3}. Every neighbour $z \in W$ of w_i lies in C_i: if z had a neighbour among $v_{i+2}, v_{i+3}, v_{i+4}$, then $d(w_i, v_{i+3}) \le 3$; if z had the neighbour v_{i-1} or v_{i+1}, then $w_i, v_i, v_{i\pm1}, z$ would be a 4-cycle. So $N(w_i) \subseteq \{v_i\} \cup C_i$.
Edges in W. Figure 3 records the two configurations used below. Let $x, y \in W$ be adjacent.

- $x \in C_0$, $y \in C_{\pm1}$: $x, v_0, v_{\pm1}, y$ is a 4-cycle.
- $x \in C_0$, $y \in C_2$: the path $w_5, v_5, v_0, x, y, v_2, v_3, w_3$ of Figure 3(a) is an induced P_8. The 6-cycle is induced, so it contributes no chord among v_5, v_0, v_2, v_3. Each off-cycle vertex meets the cycle only at its designated vertex: w_5 at v_5, x at v_0, y at v_2 and w_3 at v_3. The witness neighbourhoods, already proved, satisfy $N(w_5) \subseteq \{v_5\} \cup C_5$ and $N(w_3) \subseteq \{v_3\} \cup C_3$, so neither witness is adjacent to x, to y, or to the other witness. These three facts leave no chord.
- $x \in C_0$, $y \in C_3$: $w_1, v_1, v_0, x, y, v_3, v_4, w_4$ is an induced P_8. By symmetry, vertices of C_i and C_j with $i \neq j$ are never adjacent.
- x with trace $\{v_0, v_1\}$ and $y \in C_j$: for $j \in \{0, 1\}$ the vertices v_0, v_1, x, y span a 4-cycle; for $j = 5$ (respectively $j = 2$) the cycle x, v_0, v_5, y (respectively x, v_1, v_2, y) is a 4-cycle; for $j = 3$, $w_5, v_5, v_0, x, y, v_3, v_2, w_2$ is an induced P_8, and $j = 4$ is symmetric.
- x, y with consecutive-pair traces $\{v_0, v_1\}$ and $\{v_j, v_{j+1}\}$: $j = 0$ gives two common neighbours; $j = \pm1$ and $j = \pm2$ give 4-cycles as above; $j = 3$ gives the induced P_8 $w_2, v_2, v_1, x, y, v_4, v_5, w_5$.

The remaining paths displayed above are induced for the same three reasons, applied to the witnesses and traces named in those bullets. So every edge inside W joins two vertices of the same class C_i, and these edges form a matching. A consecutive-pair vertex is adjacent to no vertex of any C_j and to no other consecutive-pair vertex. Two vertices with the same consecutive-pair trace would have two common neighbours, so each cycle edge carries at most one such vertex, and that vertex is adjacent precisely to the two ends of the edge. Each class C_i is nonempty. This is the stated construction.
Conversely, let G be C_8, or let G be built from $C = v_0 \cdots v_5$ as in the statement. The cycle C_8 has radius 4, induced-path order 7 and no 4-cycle subgraph. Now consider the construction on C. There is no edge between distinct sets C_i, and every vertex of C_i meets C only at v_i. A cycle that meets C_i is therefore contained in $G[C_i \cup \{v_i\}]$. In that subgraph the edges are a star at v_i together with a matching, so the only cycles are the triangles formed by v_i and one matching edge.
Let y be a vertex added on the edge $\{v_j, v_{j+1}\}$, so $N(y) = \{v_j, v_{j+1}\}$. Every cycle through y uses both edges at y, and the rest of the cycle is a v_j–v_{j+1} path avoiding y. A 4-cycle through y is then a path of length two between v_j and v_{j+1} that avoids y: the two ends of the edge have a common neighbour other than y. No vertex of any C_i is adjacent to two vertices of C, and the construction places at most one vertex on each cycle edge. Those two ends therefore have no common neighbour other than the vertex added on that edge. Hence no 4-cycle passes through y. The only cycle that uses no vertex outside C is C itself. Thus G has no 4-cycle subgraph.
Let P be an induced path of order at least three. By Lemma 4.4, every vertex of P outside C is an endpoint of P, so every other vertex of P lies on C. The cycle C has no induced path on six vertices. Hence $p(G) \le 7$. Since each C_i is nonempty, one may choose $x \in C_0$ and $y \in C_2$. The path $x, v_0, v_5, v_4, v_3, v_2, y$ is induced: x and y meet the path only at v_0 and v_2, and C contributes no chord. Thus $p(G) = 7$.
There is no edge between distinct sets C_i, and a vertex added on a cycle edge has degree two on that edge, so it does not shorten distances along C. The distance from v_i to v_{i+3} is therefore three. A vertex of C_{i+3} is at distance four from v_i, and the eccentricity of v_i is at least four. It is at most four: every vertex of C_j is at distance $d_C(v_i, v_j) + 1 \le 4$ from v_i, and every vertex added on a cycle edge is within distance four of v_i. A vertex of C_i is at distance five from a vertex of C_{i+3}. A vertex y added on the edge $\{v_j, v_{j+1}\}$ has only those two neighbours. The distances from v_j and from v_{j+1} to v_{j+3} are 3 and 2, neither neighbour of y is adjacent to v_{j+3}, and $y, v_{j+1}, v_{j+2}, v_{j+3}$ is a path of length three, so a vertex of C_{j+3} is at distance four from y. Every eccentricity is at least four, and $r(G) = 4$. □

Corollary 4.5. *Let G be connected and strong C_4-free with $r(G) = 4$ and $p(G) = 7$. In the construction of Theorem 4.3, let b be the number of vertices added on edges of the 6-cycle and let m be the number of matching edges in the sets C_i. Then $S(G) = 2n - b - m$. In particular $S(G) \le 2n$, so $\ell(G) \le 2$, and* RAD(4) *holds. The statement* RAD(r) *is defined in Section 6: it asks only for $\ell \le r - 1$, and the bound just proved is stronger.*

Proof. For $G \cong C_8$ one has $n = 8, e = 8, T = 0$ and $S = 16 = 2n$, which is the formula with $b = m = 0$. In the construction, write $a = \sum_i |C_i|$. Then $n = 6 + a + b$. The edges are the six cycle edges, a edges from the sets C_i to the cycle, m matching edges and $2b$ edges incident with the vertices added on cycle edges, so $e = 6 + a + 2b + m$. Each matching edge forms one triangle with its cycle vertex, and each added vertex forms one triangle with its cycle edge, so $T = b + m$. Lemma 2.1(3) gives $S = 2e - 3T = 2n - b - m$. □

Corollary 4.6. *Let G be connected and strong C_4-free with $r(G) = 4$ and $p(G) = 7$. Then $S(G) = 2n$ if and only if $G \cong C_8$, or G is obtained from C_6 by attaching at least one leaf to each cycle vertex.*

Proof. By Corollary 4.5, $S = 2n$ if and only if $G \cong C_8$ or, in the construction, $b = m = 0$. In the latter case every vertex outside the cycle is a leaf, and each C_i is nonempty, so each cycle vertex has at least one leaf. The converse is the case $b = m = 0$ of the construction, together with C_8, both of which have $S = 2n$ and satisfy the numerical hypotheses by Theorem 4.3. □

Remark 4.7. If G is connected and strong C_4-free with $r(G) = 4$, $p(G) \ge 8$ and $\ell(G) \le 4$, then $p(G) \ge 8 = r(G) + 4 \ge r(G) + \ell(G)$. This comparison is not a structural description of G.

5 | Graphs of induced-path order at most six

In a triangle-free graph, a 4-cycle subgraph is an induced 4-cycle, and every clique has order at most two. The next classification is Lemma 3.4 of Chen and Xu [4]. Their family $\mathfrak{F} = \{F_0, F_0', F_1, \dots, F_{12}\}$ is defined by the constructions in the proofs of their Lemmas 3.1–3.3; F_5 is the Petersen graph, and the parameters t and t' attached to F_2 and F_0' are positive integers. The edge counts below use those constructions rather than the accompanying figure.

Lemma 5.1 (Chen and Xu [4, Lemma 3.4])**.** *Let G be a connected (P_7, C_4, C_3)-free graph with no clique cutset. If G is not complete, then $G \in \mathfrak{F}$.*

The members of $\mathfrak{F}$ admit the following descriptions, taken from those proofs.

- F_1 is C_5. The graph F_2 is obtained from a 5-cycle $v_1 \cdots v_5$ by adding stable sets A_1 and A_3 of equal positive order t, with a perfect matching between them, each vertex of A_1 adjacent only to v_1 and its match, and each vertex of A_3 adjacent only to v_3 and its match.
- On the same 5-cycle, F_3 has three further vertices inducing the path $a_3a_1a_4$, with a_i adjacent to v_i. The graph F_4 has four further vertices inducing the path $a_5a_3a_1a_4$, with the same spokes. The graph F_5 is the Petersen graph.
- F_0 is C_6. The graph F_0' is obtained from a 6-cycle $v_1 \cdots v_6$ by adding stable sets Y_1 and Y_4 of equal positive order t', with a perfect matching between them, each vertex of Y_1 adjacent only to v_1 and its match, and each vertex of Y_4 adjacent only to v_4 and its match.
- F_6 is C_7. On a 7-cycle $v_1 \cdots v_7$, write x_i for a vertex with neighbourhood $\{v_i, v_{i+3}\}$ on the cycle. Then F_7 is the cycle plus x_1; F_8 is the cycle plus nonadjacent x_1, x_2; F_9 is F_8 plus a vertex adjacent only to x_1 and x_2; F_{10} is the cycle plus nonadjacent x_1, x_5; F_{11} is the cycle plus a stable set $\{x_1, x_2, x_5\}$; and F_{12} is F_{11} plus a vertex adjacent only to x_1 and x_2.

Chen and Xu record that F_0, F_1, the graph F_2 with $t = 1$, F_3 and F_4 are induced subgraphs of the Petersen graph, and that $F_6, \dots, F_{11}$ are induced subgraphs of F_{12}. The graphs $F_6, \dots, F_{12}$ are the extensions of C_7 by vertices of cycle-distance three. The graphs F_2 and F_0' are the matching extensions of one pair of opposite vertices on C_5 and on C_6.

Lemma 5.2. *Every graph in $\mathfrak{F}$ other than the Petersen graph satisfies $2e \le 3n - 3$. The Petersen graph satisfies $2e = 3n$.*

Proof. The constructions give the following orders and sizes. For F_2 and F_0' the parameters t, t' are at least one.

Graph	n	e	$2e$	$3n-2e$
$F_1 = C_5$	5	5	10	5
F_2	$5+2t$	$5+3t$	$10+6t$	5
F_3	8	10	20	4
F_4	9	12	24	3
F_5	10	15	30	0
$F_0 = C_6$	6	6	12	6
F_0'	$6+2t'$	$6+3t'$	$12+6t'$	6
$F_6 = C_7$	7	7	14	7
F_7	8	9	18	6
F_8, F_{10}	9	11	22	5
F_9, F_{11}	10	13	26	4
F_{12}	11	15	30	3

The five edges of F_3 beyond C_5 are the three spokes and the two edges of $a_3a_1a_4$. The seven edges of F_4 beyond C_5 are the four spokes and the three edges of $a_5a_3a_1a_4$. Each vertex x_i contributes two edges, and each vertex placed on a pair x_ix_j contributes two further edges. The Petersen graph is the only row with difference zero. □

Lemma 5.3. *Let F be connected and strong C_4-free, and let s be a vertex at which no induced P_4 starts. Write $m = e(F[N(s)])$. Then $S(F) \le 2(|F|-1) - m$.*

Proof. A vertex at distance 3 from s would end a geodesic, hence an induced P_4, starting at s. So $V = \{s\} \cup N_1 \cup N_2$ with N_i the i-th distance layer. Each $y \in N_2$ has a unique parent $a \in N_1$. A neighbour $z \in N_2$ of y with a different parent would give the induced path $sayz$. So y has at most one neighbour in N_2, a sibling (two siblings would be common neighbours of a and y), and $N(y)$ is a clique; thus $\lambda(y) = 1$. For $a \in N_1$ we have $N(a) \subseteq \{s, \bar{a}\} \cup C_a$, where C_a is the set of children of a and $\bar{a}$ is the neighbour of a in N_1, if any. Since s and $\bar{a}$ lie in one unit, $\lambda(a) \le 1 + |C_a|$. Finally $\lambda(s) = |N_1| - m$. Summing, $S \le (|N_1| - m) + (|N_1| + |N_2|) + |N_2| = 2(|F|-1) - m$. □

Lemma 5.4. *Let G be a connected triangle-free strong C_4-free P_7-free graph of order n. Then G is obtained from a connected triangle-free strong C_4-free P_7-free graph Q with no clique cutset by attaching trees at single vertices. The core Q is K_1, K_2, or a graph in the list of Lemma 5.2. Writing $h = |V(Q)|$ and $\eta(Q) = 3h - 2e(Q)$, one has $e(G) = e(Q) + (n-h)$ and*

$$3n - 2e(G) = \eta(Q) + n - h.$$

Deleting the vertices of one attached tree, other than its root on Q, deletes exactly as many edges as vertices.

Proof. If G itself has no cut vertex, Lemma 2.10 gives that G has no clique cutset. The hypotheses of that lemma are the present hypotheses together with the absence of a cut vertex. Lemma 5.1 then applies. Its hypotheses are that G is connected, that G has no induced P_7, that G has no induced C_4 and no triangle, and that G has no clique cutset. Connectedness and the absence of a clique cutset hold by the case and by Lemma 2.10. The graph is P_7-free by hypothesis. A strong C_4-free graph has no induced C_4, and a triangle-free graph has no C_3. If G is not complete, Lemma 5.1 places it in the family $\mathfrak{F}$, hence on the list of Lemma 5.2. The only complete triangle-free graphs are K_1 and K_2. In every subcase one may take $Q = G$. The identities are then immediate.

Suppose s is a cut vertex. Path concatenation supplies a component C of $G - s$ such that every induced path of $G[C \cup \{s\}]$ that starts at s has order at most three. The first distance layer of this side is independent, since an edge there would form a triangle with s. There is no vertex at distance three or more from s, for such a vertex would end an induced P_4 at s. A vertex of the second layer has a unique parent: a second parent, together with s, would span a 4-cycle. It has no neighbour in its layer. An edge joining vertices of distinct parents is the induced path of order four recorded in the proof of Lemma 5.3, and an edge joining two children of the same parent forms a triangle with that parent. Thus $G[C \cup \{s\}]$ is a tree. Deleting the $k = |C|$ vertices of the side deletes exactly k edges. The graph that remains is connected, triangle-free, strong C_4-free and P_7-free. Iterate. The remaining graph Q has no cut vertex, and the first paragraph describes it. Reversing the deletions attaches trees at single vertices. Each deleted vertex accounted for one edge, so $e(G) = e(Q) + (n-h)$, and the displayed identity follows. □

Theorem 5.5. *Let G be a connected triangle-free strong C_4-free P_7-free graph. Then $2e(G) \le 3|V(G)|$, with equality if and only if G is the Petersen graph. In particular every connected triangle-free strong C_4-free graph of induced-path order at most 6 satisfies GEN_3 in the sense of Section 6, and so does every connected bipartite strong C_4-free graph of induced-path order at most 6. A bipartite graph in the class has strict inequality.*

Proof. Lemma 5.4 gives a core Q of order h and the identity

$$3n - 2e(G) = \eta(Q) + n - h.$$

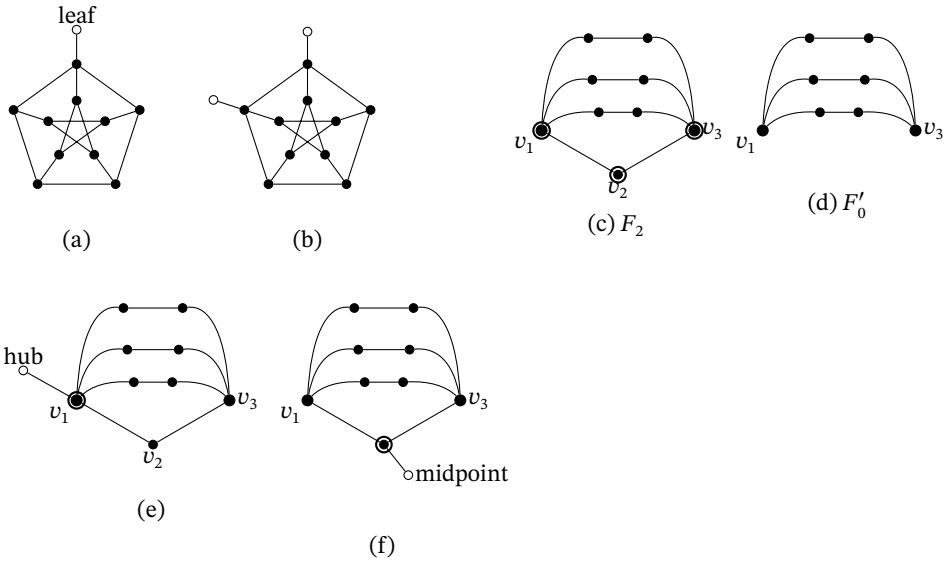
Lemma 5.2 gives $\eta(Q) \geq 0$ for every graph on its list, with equality only for the Petersen graph. The cores K_1 and K_2 have deficits 3 and 4. The right-hand side is therefore nonnegative, and it vanishes if and only if $\eta(Q) = 0$ and $n = h$. Thus equality holds if and only if G is the Petersen graph.
A bipartite graph is triangle-free. The Petersen graph is not bipartite, so the inequality is strict for every connected bipartite strong C_4-free P_7-free graph. □

Lemma 5.1 is the core classification of Chen and Xu: the connected graphs in the class with no clique cutset. The next theorem does not reclaim that list. What is new is the attachment of trees at single vertices of a core, the comparison of the resulting deficits, and the equality classification for every order $n \geq 10$. Figure 4 draws the families that meet the minimum.

Theorem 5.6. *Let $f(n)$ be the maximum of $2e(G)$ over connected triangle-free strong C_4-free P_7-free graphs of order n. For every $n \geq 10$,*

$$f(n) = 3n - \min\{n - 10,\ 5 + \mathbf{1}_{\{n \text{ even}\}}\}.$$

Write $\mathcal{P}(n)$ for the isomorphism classes of graphs obtained from the Petersen graph by placing $n - 10$ leaves on a single vertex or on the two endpoints of a single edge. For $10 \leq n \leq 14$ the extremal graphs are exactly the classes in $\mathcal{P}(n)$. For $n = 15$ they are the classes in $\mathcal{P}(15)$ together with F_2, four classes in all. For $n = 16$ they are the classes in $\mathcal{P}(16)$, the graph F_0', and the two graphs obtained from the order-15 graph F_2 by adding a leaf at a hub or at the midpoint of its unique 2-edge branch, seven classes in all. For odd $n \geq 17$ the unique extremal graph is F_2. For even $n \geq 18$ there are exactly three: F_0', and the two graphs obtained from the order-$(n-1)$ graph F_2 by adding a leaf at a hub or at the midpoint of its unique 2-edge branch.



n	Equality graphs	Classes
15	$\mathcal{P}(15)$: leaf types $(5,0)$, $(4,1)$, $(3,2)$	3
15	F_2 with $t = 5$	1
16	$\mathcal{P}(16)$: $(6,0)$, $(5,1)$, $(4,2)$, $(3,3)$	4
16	F_0'; F_2 plus a leaf at a hub; F_2 plus a leaf at the midpoint	3

FIGURE 4 | Equality families of Theorem 5.6. (a) and (b) Leaves of $\mathcal{P}(n)$ at one vertex, or at both ends of one edge. Both have deficit $n - 10$. (c) F_2: length-3 paths on distinct arcs, and one length-2 path through v_2. Deficit 5. Rings mark both hubs and the midpoint. (d) F_0': every hub-to-hub path has length 3, on its own arc. Deficit 6. (e) and (f) The allowed leaf sites, a hub and the short-branch midpoint. The table lists the four classes at order 15 and the seven at order 16.

Proof. Write $\eta(H) = 3|V(H)| - 2e(H)$ for a triangle-free graph H.
Reduction to one core. Lemma 5.4 supplies a core Q of order h and the identity $3n - 2e(G) = \eta(Q) + n - h$.
Upper bound. Every fixed row of the table in Lemma 5.2 satisfies $\eta \geq h - 10$. Each such graph has order at most 10 except F_{12}, which has order 11 and $\eta = 3$, and $3 \geq 11 - 10$. The same inequality holds for K_1 and K_2, with deficits 3 and 4. For every such core, $\eta(Q) + n - h \geq n - 10$. The two parameterised rows are F_2, of odd order and deficit 5, and F_0', of even order and deficit 6. If the core is F_2 and the final order is even, then at least one vertex has been attached, so the displayed

deficit is at least 6. If the core is F_2 and the final order is odd, the deficit is at least 5. If the core is F_0', the deficit is at least 6. In every case

$$3n - 2e(G) \geq \min\{n - 10,\ 5 + \mathbf{1}_{\{n \text{ even}\}}\}.$$

Attainment. The value $f(n)$ is attained by the graphs named in the statement; the next paragraph shows that no other graph attains it. A leaf added to a triangle-free strong C_4-free graph creates neither a triangle nor a cycle. Realise the Petersen graph on the 2-element subsets of $\{1, 2, 3, 4, 5\}$, with two subsets adjacent when they are disjoint. This is the graph P of Theorem 3.6. The symmetric group on the ground set acts on the vertices. The action is vertex-transitive and edge-transitive, and it carries any two nonadjacent vertices to $\{1, 2\}$ and $\{1, 3\}$. The path

$$\{1, 2\}, \{3, 4\}, \{1, 5\}, \{2, 3\}, \{1, 4\}$$

is an induced P_5 starting at $\{1, 2\}$, and

$$\{1, 2\}, \{3, 4\}, \{1, 5\}, \{2, 4\}, \{1, 3\}$$

is an induced P_5 with ends $\{1, 2\}$ and $\{1, 3\}$. Thus every vertex of P is an end of an induced P_5, and every pair of nonadjacent vertices is the pair of ends of an induced P_5.

Place a leaves at one end of an edge of P and b leaves at the other, with $a + b = n - 10$ and $a \geq b \geq 0$. The case $b = 0$ is the placement at a single vertex. The graph is connected, triangle-free and strong C_4-free. It has $15 + (n - 10)$ edges, so its deficit is $n - 10$. An induced path uses a leaf only as an end. A path that uses two leaves at the same vertex has order three. A path that uses one leaf at each end of the chosen edge is the edge itself with the two leaves attached: the ends of that edge are adjacent, so no longer induced path of P has those ends. A path that uses one leaf restricts to an induced path of P and has order at most six, because P is P_6-free. Thus the graph is P_7-free. Vertex-transitivity and edge-transitivity give one isomorphism class for each pair (a, b). Distinct pairs give nonisomorphic graphs: the Petersen graph is cubic, so the degrees of the vertices adjacent to a leaf are $3 + a$ and $3 + b$ when $b \geq 1$, and the single value $3 + a$ when $b = 0$. These are the classes in $\mathcal{P}(n)$, and they attain deficit $n - 10$.

For the deficit $5 + \mathbf{1}_{\{n \text{ even}\}}$, use F_2 at odd orders and F_0' at even orders. In the descriptions preceding Lemma 5.2, each of these graphs consists of internally disjoint three-edge paths between two hubs, together with one additional two-edge path in the case of F_2. Every cycle uses two of those paths and therefore has length at least five. An induced path that contains at most one hub meets that hub in at most two of the paths and so has at most five vertices. An induced path that contains both hubs uses at most one path between them, a segment of at most three edges, and each end extends by at most one further edge before the next vertex meets the opposite hub and creates a chord. Its order is at most six. The table of Lemma 5.2 gives deficits 5 and 6 respectively. The smaller of the two deficits is therefore attained.

Equality. Write $\delta(n)$ for the minimum in the statement, and let G satisfy $3n - 2e(G) = \delta(n)$. The identity $3n - 2e(G) = \eta(Q) + n - h$ and the upper bound force the core Q to be the Petersen graph, F_2 or F_0'. Every other core in the table of Lemma 5.2, and each of K_1 and K_2, has $\eta(Q) - h \geq -8$, hence deficit at least $n - 8$. For $n \geq 10$ one has $n - 8 > \delta(n)$, since $\delta(n) \leq n - 10$.

If Q is the Petersen graph, then $\eta(Q) = 0$, $h = 10$ and the deficit is $n - 10$. This equals $\delta(n)$ if and only if $10 \leq n \leq 16$. The vertices outside Q form trees attached at single vertices of Q. A tree of depth at least two contains a path y, a, s of length two ending at its attachment vertex s. Concatenating that path with an induced P_5 of Q that starts at s produces an induced P_7: the only edge from $\{y, a\}$ into Q is as, and a is not adjacent to the neighbour of s on the P_5 because that neighbour and a would form a triangle with s. Thus every attached tree is a star of leaves at its attachment vertex.

The set A of vertices of Q that are adjacent to at least one leaf is a clique. Suppose $u, v \in A$ are distinct and nonadjacent, and let ℓ_u and ℓ_v be leaves at u and at v. The paragraph on attainment supplies an induced P_5 of Q with ends u and v; write it $u = q_0, q_1, q_2, q_3, q_4 = v$. The path $\ell_u, q_0, q_1, q_2, q_3, q_4, \ell_v$ has seven vertices. It is induced. The leaf ℓ_u is adjacent only to q_0, hence to none of $q_1, q_2, q_3, q_4, \ell_v$. The leaf ℓ_v is adjacent only to q_4, hence to none of $\ell_u, q_0, q_1, q_2, q_3$. The path $q_0q_1q_2q_3q_4$ is induced, so it contributes no chord. This is an induced P_7, which is forbidden. Thus no two vertices of A are nonadjacent, and A is a clique. The Petersen graph is triangle-free, so a clique has size at most two. If $|A| = 2$, the two vertices are adjacent. The attachments therefore occupy a single vertex or the two ends of one edge. These are the classes in $\mathcal{P}(n)$.

If $Q = F_2$, then $\eta(Q) = 5$ and $h = 5 + 2t$ is odd, so the deficit is $5 + n - h$. For odd n this equals $\delta(n) = 5$ if and only if $n = h$ and $n \geq 15$, in which case G is F_2. For even n the orders n and h have opposite parity, so $n - h \geq 1$ and the deficit is at least 6. Equality with $\delta(n)$ holds if and only if $n - h = 1$ and $\delta(n) = 6$, hence $n \geq 16$ is even and G is F_2 of order $n - 1$ with a single leaf. Here $t = (n - 6)/2 \geq 5$.

If $Q = F_0'$, then $\eta(Q) = 6$ and the deficit is $6 + n - h \geq 6$, with equality if and only if $n = h$. Then $\delta(n) = 6$, so $n \geq 16$ is even and G is F_0'.

It remains to decide which vertices of F_2 may carry the leaf. Adding a leaf at a vertex v of a P_7-free graph preserves P_7-freeness if and only if v is not an end of an induced P_6: the leaf has degree one, so it is an end of every induced path that

contains it, and the remainder is an induced path starting at v. Label F_2 as in the description before Lemma 5.2. The vertex set consists of the hubs v_1 and v_3, the midpoint v_2 of the unique 2-edge branch $v_1v_2v_3$, and the internal vertices of the long branches. The long branches are the t matching paths through $A_1 \cup A_3$ and the path $v_1v_5v_4v_3$. The map that swaps v_1 with v_3, swaps v_5 with v_4, fixes v_2, and swaps the two ends of each matching edge preserves the cycle, the matching and every spoke, so it is an automorphism. It exchanges the two hubs.

No induced P_6 starts at a hub. Let P be an induced path starting at v_1. If the second vertex is v_2, the third is v_3, and the fourth is either v_4 or a vertex of A_3. In the first subcase the only continuation is v_5, and v_1v_5 is a chord, so the induced path has order at most four. In the second the only continuation is the match in A_1, and that match is adjacent to v_1, so again the order is at most four. If instead the path leaves v_1 along a long branch, it may continue to v_3 and then to the v_3-end of one further long branch. The resulting path has order five. Its last vertex has no neighbour off that path except its partner on the same long branch, and that partner is adjacent to v_1. The path does not extend, and the edge v_1v_2 is a chord of any continuation through v_2. Thus no induced P_6 starts at v_1. The automorphism gives the same conclusion at v_3.

No induced P_6 starts at v_2. Every induced path that starts at v_2 has second vertex v_1 or v_3. From v_1 it can enter only one long branch, and the far end of that branch is v_3, which is adjacent to v_2. The path stops before v_3 and has order at most four.

Every other vertex is an end of an induced P_6. The remaining vertices are the internal vertices of the long branches, namely $A_1 \cup A_3 \cup \{v_4, v_5\}$. Let p be such a vertex and adjacent to v_1, so $p \in A_1$ or $p = v_5$. The automorphism carries these vertices to the vertices adjacent to v_3. Since $t \geq 5$, at least two long branches avoid p. Choose two matching branches v_1abv_3 and v_1cdv_3 among them, with $a, c \in A_1$ and with $b, d \in A_3$ their matches, and consider the path p, v_1, a, b, v_3, d. The edges pv_1, v_1a, ab, bv_3 and v_3d are present by construction. There is no chord. A vertex of A_1 is adjacent only to v_1 and its match, a vertex of A_3 is adjacent only to v_3 and its match, A_1 and A_3 are stable sets, and the hubs are nonadjacent. If $p \in A_1$, its match is neither b nor d, so p meets the path only at v_1. If $p = v_5$, the only neighbours of v_5 are v_1 and v_4, and v_4 is not on the path, so again p meets the path only at v_1. The vertex a meets the path only at v_1 and b, and b meets the path only at a and v_3. Thus the path is an induced P_6, and a leaf at p extends it to an induced P_7.

A leaf at a hub or at v_2 therefore keeps the graph P_7-free, and a leaf at any other vertex does not. These are the only legal positions. The two hubs are equivalent under the automorphism, while v_2 is the unique common neighbour of the hubs, so the two placements are not isomorphic. In F_2 the hub v_1 has degree $2 + t$, because it is adjacent to v_2, to v_5 and to the t vertices of A_1, and v_2 has degree 2. After the leaf is added, the neighbour of the leaf has degree $t + 3$ or degree 3 according as it is a hub or v_2. For $t \geq 5$ these degrees differ.

The graphs that meet $\delta(n)$ are pairwise nonisomorphic. They are separated by the number of vertices of degree one, together with the multiset of degrees of the vertices adjacent to a leaf. A graph with a single leaf is separated by the degree of the neighbour of that leaf. For a pair (a, b) those degrees are $3 + a$ and, when $b \geq 1$, $3 + b$. The graphs F_2 and F_0' have minimum degree two. On a one-leaf extension of F_2 the neighbour of the leaf has degree $t + 3$ or degree 3, and these differ for $t \geq 5$. The order-15 core is F_2 with $t = 5$. At order 16 the underlying core has order 15, so that degree is 8 or 3. At even order $n \geq 18$ the underlying core is F_2 of order $n - 1$, and the degree is $n/2$ or 3. Table 2 is the resulting count. The order-15 and order-16 rows are the families drawn in Figure 4. □

n	Equality graphs	Leaves	Degrees at the leaf support	Classes
$10 \leq n \leq 14$	$\mathcal{P}(n)$, pairs (a, b)	$n - 10$	none ($n = 10$); else $\{3 + a\}$ or $\{3 + a,\ 3 + b\}$	one per pair
15	$\mathcal{P}(15)$: $(5, 0)$, $(4, 1)$, $(3, 2)$; and F_2, $t = 5$	5 or 0	{8}, {7, 4}, {6, 5}, or none	4
16	$\mathcal{P}(16)$: $(6, 0)$, $(5, 1)$, $(4, 2)$, $(3, 3)$; F_0'; two one-leaf graphs	6, 0 or 1	{9}, {8, 4}, {7, 5}, {6, 6}; none; 8 or 3	7
odd $n \geq 17$	F_2	0	none	1
even $n \geq 18$	F_0' and the two one-leaf graphs	0 or 1	none, or $n/2$, or 3	3

TABLE 2 | Separation of the equality graphs of Theorem 5.6 by the number of leaves and the degrees of the vertices adjacent to a leaf.

Lemma 5.7. *Let G be a connected strong C_4-free P_7-free graph that contains a triangle, satisfies $L(v) \geq 4$ for every vertex, and contains no two adjacent vertices of degree two. Then G is isomorphic to the graph $H_{3,2}$ obtained by partitioning the vertices of K_5 into parts of sizes three and two and subdividing each of the six edges with one end in each part exactly once. This graph has*

$$(n, e, T, S) = (11, 16, 1, 29).$$

Proof. Let A be the part of size three, let B be the part of size two, and let Z be the set of six subdivision vertices. The construction puts a triangle on A, an edge on B, no edge between A and B, and exactly one vertex of Z on each pair with

one end in A and one end in B. Thus $n = 11$ and $e = 3 + 1 + 12 = 16$. A vertex of Z is adjacent to one vertex of A and one vertex of B, which are nonadjacent, and no vertex of Z lies in a triangle with an edge of A or with the edge of B. The only triangle is the triangle on A, so $T = 1$ and $S = 2e - 3T = 29$. Each vertex of A has degree four and lies in one triangle, so its deletion loss is 5. Each vertex of B has degree four and lies in no triangle, so its deletion loss is 8. Each vertex of Z has degree two, with nonadjacent neighbours, so its deletion loss is 4. No two vertices of Z are adjacent. Figure 5 draws the graph.

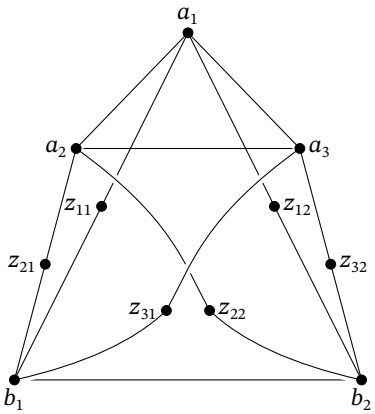


FIGURE 5 | The graph $H_{3,2}$ of Lemma 5.7. The part $\{a_1, a_2, a_3\}$ is the unique triangle, and $\{b_1, b_2\}$ is the remaining edge of K_5. The vertex z_{ij} subdivides $a_i b_j$. A white gap is a crossing with no shared vertex. The curved paths are $a_2 z_{22} b_2$ and $a_3 z_{31} b_1$.

There is no 4-cycle. The subgraph induced by $A \cup B$ is a triangle together with a disjoint edge. A cycle through a vertex $s \in Z$ uses both edges at s, so it contains the path from its neighbour in A to its neighbour in B. Those two vertices have no common neighbour other than s.

There is no induced P_7. The set A is a clique and the set B is a clique, so an induced path contains at most two vertices of A and at most two vertices of B. An induced path on seven vertices would therefore contain at least three vertices of Z. A vertex of Z on an induced path is an end of the path, or else lies between its two neighbours. Suppose two vertices of Z are internal. If their neighbours in B are distinct, those two vertices of B are adjacent, hence consecutive on the path, and the path contains a subpath $a\,s\,b\,b'\,s'\,a'$ with $a, a' \in A$. If their neighbours in B coincide, the path contains a subpath $a\,s\,b\,s'\,a'$ with $a, a' \in A$. In both cases the edge aa' is a chord. Thus at most one vertex of Z is internal. At least three vertices of Z lie on the path, and a path has only two ends, so the path contains exactly three vertices of Z: the two ends and one internal vertex. The other four vertices lie in $A \cup B$.

The internal vertex contributes a subpath asb with $b \in B$. Suppose both vertices of B lie on the path. They are consecutive, so the path contains the block $asbb'$. The fourth vertex of $A \cup B$ on the path is a vertex $a' \in A$, and $asbb'$ is a block of four consecutive vertices. There is no edge between A and B, so a' is not adjacent to b'. The only remaining possibility is that a' is adjacent to a on the path. The two ends of the resulting seven-vertex path lie in Z. The end adjacent to a' has its other neighbour in B, because its neighbour on the path lies in A. Both vertices of B are already on the path and neither is the successor of that end, so the other neighbour is a chord. Suppose instead that only one vertex of B lies on the path. Then all three vertices of A lie on the path. Those three vertices form a clique, so they do not lie on an induced path. This contradiction shows that $H_{3,2}$ is strong C_4-free and P_7-free, and the local hypotheses have already been checked. Thus $H_{3,2}$ is a graph of the kind named in the lemma.

It remains to show that no other isomorphism type occurs. Let G satisfy the hypotheses and fix an induced triangle as the starting subgraph, still called H during the growth. A labelled induced subgraph is enlarged by the following rule. If some vertex v of the current graph H has deletion loss $L_H(v) < 4$, computed in H, and v has no neighbour in G outside H, then the degree and the triangle count of v have attained their values in G, so $L_G(v) < 4$. If a and b are adjacent vertices of degree two in H, then at least one of them has a neighbour in G outside H. The rule then selects the pair $X = \{a, b\}$. It does not designate one endpoint of the pair in advance. In the deficient-vertex situation the rule selects the singleton consisting of that vertex. In either situation a new vertex required by the rule has neighbourhood U in the current vertex set satisfying $U \cap X \neq \emptyset$. When a and b are the adjacent degree-two vertices, this is the condition $U \cap \{a, b\} \neq \emptyset$. If H already satisfies $L_H \geq 4$ at every vertex and has no two adjacent vertices of degree two, its isomorphism type is compared with $H_{3,2}$. If that graph is still a proper induced subgraph of G, connectivity supplies a vertex outside H with a neighbour in H, and every nonempty neighbourhood is admitted at that node.

For the next labelled vertex z, form every set $U \subseteq V(H)$ that meets the required set X, or every nonempty U at an unrestricted node, and join z precisely to U. Keep the child when the new graph has neither a 4-cycle subgraph nor an induced P_7. A new 4-cycle exists precisely when two vertices of U have a common neighbour in the current graph. An induced P_7 created by the addition uses z and six vertices of the current graph. Appendix A of the supplement (Section A) records the equivalent test: after z is deleted, the six vertices form one path or two path components, and the neighbourhood of z contains exactly one endpoint of each path component and no other vertex of that set. Fix any embedding of the current graph into G. The neighbourhood, in the current vertex set, of a vertex that the rule requires is one of these sets U, and the child it determines is kept. An embedded proper prefix therefore extends. Every graph satisfying the hypotheses arises along some branch, with no restriction on its order.

The search tree is finite. Appendix A of the supplement, Section A, records the audit. The same certificate is checked by the submission program `certificates/gen3/verify_gen3.py`. From the repository root the command is `python3 certificates/gen3/verify_gen3.py`. The tree has 169 labelled nodes and 52 leaves. By order $3, \dots, 17$ the numbers of visited nodes are

$$1, 1, 1, 1, 2, 3, 5, 14, 23, 24, 23, 19, 24, 20, 8.$$

Four nodes are unrestricted. All four have order 11, and each is the graph $H_{3,2}$. The search continues to expand $H_{3,2}$ from those nodes. At an unrestricted node every nonempty neighbourhood is admitted, and a child is kept when it creates neither a 4-cycle subgraph nor an induced P_7. A later node at which a local condition fails is kept in the tree. Expansion continues, and every new neighbourhood is required to meet the deficient vertex or the adjacent pair of degree-two vertices. Every branch terminates. The largest graph order reached by the search is 17. The verifier imposes no order cutoff and performs no isomorphism pruning. No node other than the four copies of $H_{3,2}$ satisfies both local conditions. Therefore $G \cong H_{3,2}$. □

Theorem 5.8. *Every connected strong C_4-free P_7-free graph G satisfies $S(G) \leq 3n$, with equality if and only if G is the Petersen graph.*

Proof. The triangle-free case, including the equality, is Theorem 5.5. Suppose G contains a triangle and $S(G) \geq 3n$, and choose such a connected graph of minimum order. Every smaller connected graph in the class satisfies $S \leq 3|V|$, and the inequality is strict whenever the smaller graph contains a triangle.

Suppose s is a cut vertex. Induced paths that start at s in distinct components of $G - s$ concatenate. Since $p(G) \leq 6$, some component determines a side G_1, induced by that component together with s, in which every induced path starting at s has order at most three. Let G_2 be the subgraph induced by the remaining vertices together with s, and write $n_i = |V(G_i)|$. Then $n = n_1 + n_2 - 1$ and $n_1 \geq 2$. Lemma 5.3 gives $S(G_1) \leq 2(n_1 - 1)$. The graph G_2 is a smaller connected graph in the class. If it contains a triangle, minimality gives $S(G_2) < 3n_2$; if it does not, Theorem 5.5 gives $S(G_2) \leq 3n_2$. Lemma 2.3 therefore yields

$$S(G) \leq 2(n_1 - 1) + 3n_2 = 3n - (n_1 - 1) < 3n.$$

Thus G has no cut vertex, and $G - v$ is connected for every vertex v.

Suppose $L(v) \leq 3$ for some vertex v. Then $S(G - v) = S(G) - L(v) \geq 3n - 3 = 3(n - 1)$. The graph $G - v$ is connected and lies in the class. It contains no triangle: otherwise minimality would give the strict inequality $S(G - v) < 3(n - 1)$. Theorem 5.5 therefore forces $G - v$ to be the Petersen graph, so $n = 11$ and $S(G - v) = 30$. Any two nonadjacent vertices of the Petersen graph have a common neighbour, so the neighbours of v form a clique. The Petersen graph is triangle-free, and a clique in it has size at most two. The graph G contains a triangle, so v is adjacent to both ends of an edge and to no further vertex of $G - v$. Then $d(v) = 2$, $t(v) = 1$ and $L(v) = 2 \cdot 2 - 3 \cdot 1 = 1$, hence $S(G) = 31 < 33$. This contradiction shows that $L(v) \geq 4$ for every vertex v.

Suppose u and w are adjacent vertices of degree two. The bound on the deletion loss forces $t(u) = t(w) = 0$, so neither vertex lies in a triangle and the two neighbours of u are nonadjacent. Let x be the neighbour of u other than w, and let y be the neighbour of w other than u. Then $x \neq y$, the edge xy is absent, and x is not adjacent to w. Deleting $\{u, w\}$ removes the three edges xu, uw and wy and preserves every triangle. The graph that remains may be disconnected. Its number of edges is $e(G) - 3$ and its number of triangles is $T(G)$, so its value of S is $S(G) - 6 \geq 3n - 6 = 3(n - 2)$. For a disconnected graph, S is the sum of the values on the components. Each component is a smaller connected graph in the class. The triangle of G survives in one component, and minimality makes the inequality strict for that component, while every other component contributes at most three times its order. The sum is therefore strictly less than $3(n - 2)$, a contradiction.

Thus G satisfies the hypotheses of Lemma 5.7. That lemma gives $n = 11$ and $S(G) = 29 < 33$, which is impossible. No connected graph in the class with a triangle satisfies $S \geq 3n$. Combined with Theorem 5.5, this is the stated bound, and equality holds only for the Petersen graph. □

Theorem 5.9. *Let G be a connected graph on at least two vertices with $\ell(G) \le 4$. Then*

$$p(G) \ge r(G) + \ell(G)^{c_{C_4}(G)}.$$

If $c_{C_4}(G) = 0$, the power equals 1. The conjecture remains open for graphs with $\ell > 4$.

Proof. If G contains a 4-cycle, then $c_{C_4}(G) = 0$ and the inequality is $p(G) \ge r(G)+1$. A shortest path between two vertices at distance $\mathrm{diam}(G) \ge r(G)$ is induced.
Suppose now that G is strong C_4-free, and write $r = r(G)$. The inequality is $p \ge r + \ell$.

- $r = 1$. Lemma 3.1 gives $\ell < 2$ and $p \ge r + \lceil \ell \rceil$, hence $p \ge r + \ell$.
- $r = 2$. If $\ell \le 2$, Lemma 3.2 gives $p \ge 4$. If $2 < \ell \le 3$ and $p \le 4$, Lemma 3.3 gives $\ell \le 2$, a contradiction, so $p \ge 5$. If $3 < \ell \le 4$ and $p \le 5$, Theorem 3.6 gives $\ell \le 3$, a contradiction, so $p \ge 6$.
- $r = 3$. If $\ell \le 2$, Theorem 2.4 gives $p \ge 5$. If $2 < \ell \le 3$ and $p \le 5$, then Lemma 3.5 and $r = 3$ give $G = C_6$, with $\ell = 2$, so $p \ge 6$. Theorem 3.6 gives $\ell \le 3$ for every P_6-free graph in the class, which does not rule out $\ell = 3$. If $3 < \ell \le 4$, a failure of $p \ge r + \ell$ would force $p \le 6$, and Theorem 5.8 gives $\ell \le 3$.
- $r = 4$. Theorem 2.4 gives $p \ge 7$. If $p = 7$, Corollary 4.5 gives $\ell \le 2$, so $p \ge r + \ell$. If $p \ge 8$, Remark 4.7 gives the same inequality.
- $r \ge 5$. Theorem 2.4 gives $p \ge 2r - 1 \ge r + 4 \ge r + \ell$.

The comparison does not reach $\ell > 4$. Corollary 6.1 and Section 7 record what remains. □

Equality in REAL-133 occurs for C_5 ($r = 2$, $\ell = 2$, $p = 4$), the Petersen graph ($r = 2$, $\ell = 3$, $p = 5$) and C_6 ($r = 3$, $\ell = 2$, $p = 5$). We do not characterise all equality cases.

Remark 5.10. Every connected graph G with $\ell(G) < 5$ satisfies

$$p(G) \ge r(G) + \left\lfloor \ell(G)^{c_{C_4}(G)} \right\rfloor.$$

The exponent follows the convention of Conjecture 1.1: if $c_{C_4}(G) = 0$, then the power equals 1. If G contains a 4-cycle, both the real form and this floor form reduce to $p \ge r + 1$, which follows from a geodesic. If G is strong C_4-free and $\ell \le 4$, Theorem 5.9 gives $p \ge r + \ell$, and $\ell \ge \lfloor \ell \rfloor$. Suppose G is strong C_4-free and $4 \le \ell < 5$, and suppose the floor form fails. Then $\lfloor \ell \rfloor = 4$, so $p \le r + 3$. Theorem 2.4 gives $p \ge 2r - 1$, so $r \le 4$. If $r \le 3$, then $p \le 6$, and Theorem 5.8 gives $\ell \le 3$, contradicting $\ell \ge 4$. If $r = 4$, then $p = 7$, and Corollary 4.5 gives $\ell \le 2$, again a contradiction. The real-valued conjecture for $\ell > 4$ remains open.

6 | The reduction above density four

For integers $K \ge 3$ and $r \ge 4$ consider the following statements.

GEN$_K$: every connected strong C_4-free graph G with $p(G) \le 2K$ satisfies $S(G) \le Kn$.

RAD(r): every connected strong C_4-free graph with $r(G) = r$ and $p(G) = 2r - 1$ satisfies $\ell(G) \le r - 1$.

Corollary 4.5 is the case $r = 4$ of RAD(r), in the stronger form $\ell \le 2$.

Corollary 6.1. *Let G be a connected strong C_4-free graph that violates Conjecture 1.1. Then $\ell(G) > 4$. Put $K = \lceil \ell(G) \rceil - 1 \ge 4$. Then either*
(a) $r(G) \le K$ and $p(G) \le r(G) + K \le 2K$, so G violates GEN$_K$*; or*
(b) $r(G) = K + 1 \ge 5$ and $p(G) = 2r(G) - 1$, and $\ell(G) > r(G) - 1$, so G violates RAD($K + 1$).

Proof. Write $r = r(G)$, $p = p(G)$ and $c = \lceil \ell(G) \rceil = K + 1$. Then $\ell > 4$ by Theorem 5.9, so $K \ge 4$. A violation means $p \le r + c - 1 = r + K$. Theorem 2.4 gives $2r - 1 \le p \le r + K$, so $r \le K + 1$. If $r \le K$, then $p \le r + K \le 2K$, and $S > Kn$ because $\ell > K$. If $r = K + 1$, then $2r - 1 \le p \le 2K + 1 = 2r - 1$, and $\ell > K = r - 1$. The radius in (b) is at least 5. □

Since GEN$_K$ ignores the radius, it is stronger than alternative (a). The statement GEN$_K$ might fail for large K even if REAL-133 holds; see Remark S2.5.

Proposition 6.2. *For $k \geq 3$ let θ_k consist of two vertices x, y joined by k internally disjoint paths xa_ib_iy. Then θ_k is strong C_4-free, $n = 2k + 2$, $r = 3$, $p = 6$ and $S = 6k = 3n - 6$. Thus $p - r - \ell = 6/n \to 0$.*

Proof. Every cycle consists of two complete branches and so has length 6; in particular there is no 4-cycle. There are no triangles, so $S = 2e = 6k$. Every vertex has eccentricity 3: $d(x, y) = 3$ and $d(a_i, b_j) = 3$ for $i \neq j$. The path $a_jxa_ib_iyb_l$ (with j, i, l distinct) is induced. Conversely, let P be an induced path. If P avoids x and y, it lies inside one edge a_ib_i. If P contains x but not y, then x has at most two neighbours on P, each followed on P by at most its b-vertex, so $|P| \leq 5$. If P contains x and y, then every a_j on P is adjacent on P to x and every b_j to y, at most one branch lies entirely on P, and so P consists of x, y, at most one full branch and at most one further neighbour of each of x and y. Hence $|P| \leq 6$. □

Corollary 6.3. *Let G be a connected strong C_4-free graph with $p(G) = 6$. Then $S(G) \leq 3n - 1$ and $\ell(G) < 3$. On each of the classes $(r, p) = (2, 6)$ and $(r, p) = (3, 6)$ the supremum of ℓ is 3, and the supremum is not attained.*

Proof. The graph is P_7-free. Theorem 5.8 gives $S(G) \leq 3n$, with equality only for the Petersen graph. The Petersen graph has induced-path order 5, so $S(G) \leq 3n - 1$ and $\ell(G) < 3$.
Both constructions below are strictly less than 3 and tend to 3 as the order tends to infinity. The supremum on each class is therefore 3, and it is not attained.
For the class $(2, 6)$, take the graph F_2 of Theorem 5.6 with parameter $t \geq 2$. In the description preceding Lemma 5.2 the hubs are v_1 and v_3, the unique branch of length 2 is $v_1v_2v_3$, and the remaining v_1–v_3 paths are the t matching paths through $A_1 \cup A_3$ together with $v_1v_5v_4v_3$. These $t + 1 \geq 3$ paths are branches of length 3. The order is $n = 5 + 2t$. The graph is triangle-free, and the table of Lemma 5.2 gives $2e = 3n - 5$, so $\ell = 3 - 5/n$. Every vertex other than the two hubs is adjacent to a hub, and v_2 is adjacent to both hubs, so every vertex is within distance 2 of v_2. A vertex of A_1 is not adjacent to v_2. Each hub has degree $t + 2 = (n - 1)/2 < n - 1$, and every other vertex has degree 2, so no vertex dominates the graph. The radius is 2. The proof of Theorem 5.6 shows that every induced path has order at most 6. For the matching lower bound, choose three branches of length 3, written v_1abv_3, v_1cdv_3 and v_1efv_3. The path a, v_1, c, d, v_3, f is induced: the hubs are nonadjacent, and an internal vertex of any one of these branches is adjacent only to its two neighbours on that branch. Thus $(r, p, \ell) = (2, 6, 3 - 5/n)$.
For the class $(3, 6)$, Proposition 6.2 gives the graph θ_k for every $k \geq 3$, with $\ell = 3 - 6/n$. □

The two families of Corollary 6.3 are why a uniform gap below 3 is not available. Section 7 is the boundary of the argument.

7 | What is proved and what is open

Table 3 is the boundary between the proved range $\ell \leq 4$ and the open range $\ell > 4$. The comparisons with earlier work are in the introduction and in Section 5. The dependency note in the supplement is the only list of which supplement arguments are hypotheses of a theorem here. A finite search is a proof here only in the form fixed for Lemma 5.7, and for the two-triangle search of Lemma S5.3 in the supplement.

Statement	Status
$M(n)$ for connected strong C_4-free P_6-free graphs, with uniqueness for $n \geq 7$	proved, Theorem 3.6
graphs with $r = 4$ and $p = 7$, the formula $S = 2n - b - m$, and the equality family	proved, Theorem 4.3 and Corollaries 4.5, 4.6
$2e \leq 3n$ for connected triangle-free strong C_4-free P_7-free graphs, equality only for the Petersen graph	proved, Theorem 5.5
the exact value $f(n)$ and the extremal graphs, $n \geq 10$	proved, Theorem 5.6
$S \leq 3n$ for every connected strong C_4-free P_7-free graph, equality only for the Petersen graph	proved, Theorem 5.8
if $p = 6$, then $S \leq 3n - 1$ and $\ell < 3$; the supremum 3 is sharp and unattained for $(2, 6)$ and for $(3, 6)$	proved, Corollary 6.3
REAL-133 for every connected graph with $\ell \leq 4$	proved, Theorem 5.9
REAL-133 for $\ell > 4$, and REAL-133 with no restriction on ℓ	open

TABLE 3 | Proved statements for $\ell \leq 4$, and what remains open.

The method stops at $\ell = 4$ because that is the last density closed by the structure already proved. For a strong C_4-free graph, Theorem 3.6 gives $\ell \leq 3$ whenever $p \leq 5$, Theorem 5.8 gives $\ell \leq 3$ whenever $p \leq 6$, and Corollary 4.5 gives $\ell \leq 2$ when $r = 4$ and $p = 7$. Theorem 2.4 gives $p \geq 2r - 1$, hence $p \geq r + 4$ once $r \geq 5$ and $\ell \leq 4$. These comparisons are the

cases in the proof of Theorem 5.9. A graph with $\ell > 4$ has $p \geq 7$, so the bound for induced-path order at most 6 does not apply. If its radius is at least 5, the radius-four classification does not apply either.
The next obstacle is the alternative in Corollary 6.1: either $S > Kn$ for some $K \geq 4$ on a connected strong C_4-free graph with $p \leq 2K$, or a connected strong C_4-free graph of radius $r \geq 5$ and induced-path order $2r-1$ with $\ell > r-1$. No structure theorem for that radius is proved here. Lemma S4.2 gives $\ell < 4$ at induced-path order 7 only in the triangle-free case. The supplement's triangle-free graph with parameters $(r, p, n, S, \ell) = (2, 7, 12, 34, 17/6)$ shows that induced-path order 7 does not by itself force a triangle, and the supplement records the pairs that remain in the bands $4 < \ell \leq 5$ and $5 < \ell \leq 6$.
Three constructions already in the text block the naive extensions. The graphs F_2 and θ_k of Corollary 6.3 have induced-path order 6 and densities $3-5/n$ and $3-6/n$, so there is no uniform gap below 3 from which a longer induced path would inherit $S \leq 3n$. Example S3.1 gives, for every radius $r \geq 6$, a strong C_4-free graph with $p = 2r-1$ for which $e = n+T$ fails, so the radius-four count does not extend by that identity. Example S3.2 shows that a uniform lower bound on the deletion loss does not bound λ. Remark S2.5 records that a Moore graph of degree 57, if it exists, satisfies REAL-133, while a proof of every mass bound GEN_K is a different assertion.

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